Closed-Form Richardson-Number Stability Functions from Monin–Obukhov Similarity Theory: Re-Derivation, Generalization, and Implementation Notes

A Technical Working Reference for Boundary-Layer Modeling

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Abstract

This document provides a mathematically rigorous, fully self-consistent reference for transforming Monin–Obukhov Similarity Theory (MOST) flux-profile functions $\phi_m(\zeta)$ and $\phi_h(\zeta)$ into explicit, closed-form gradient Richardson number (Ri) stability functions $S_m(Ri)$ and $S_h(Ri)$. We correct historical transcription errors in the $\zeta \leftrightarrow Ri$ mapping relation, generalize the stable boundary layer (SBL) formulation to arbitrary heat profile exponents α_h, detail the unstable branch solutions, derive near-neutral series expansions via series reversion, and outline numerical implementation considerations (including surface roughness z_0 integration). This working reference is designed for boundary-layer researchers and numerical modelers at UAH.

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1 Theoretical Foundations & Coupling Identicalities

In first-order eddy-diffusivity closures (K-theory), turbulent exchange coefficients for momentum (K_m) and heat (K_h) are parameterized using mixing-length theory driven by mean vertical shear $S = |\partial u/\partial z|$:

$$K_m = l_m^2 S S_m(Ri), \qquad K_h = l_h^2 S S_h(Ri) \tag{1}$$

where $l_m = l_h = \kappa z$ near the surface. Conversely, Monin–Obukhov Similarity Theory (MOST) defines diffusivities as:

$$K_m = \frac{\kappa z u_*}{\phi_m(\zeta)}, \qquad K_h = \frac{\kappa z u_*}{\phi_h(\zeta)} \tag{2}$$

Equating K-theory with MOST using the surface-layer profile relation $S = \frac{u_*}{\kappa z}\phi_m(\zeta)$ yields the fundamental coupling definitions:

$$S_m(Ri) = \frac{1}{\phi_m^2(\zeta)} \tag{1a}$$

$$S_h(Ri) = \frac{1}{\phi_m(\zeta)\phi_h(\zeta)} = \frac{S_m^{1/2}(Ri)}{\phi_h(\zeta)} \tag{1b}$$

1.1 The Corrected $\zeta \leftrightarrow Ri$ Transformation

The gradient Richardson number Ri is defined as:

$$Ri = \frac{g}{\theta_0}\frac{\frac{\partial\theta}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2} = \zeta\frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \tag{3}$$

Substituting (1a) and (1b) into (3) yields the exact relation connecting Ri and ζ:

$$Ri = \zeta\frac{S_m^{1/2}/S_h}{S_m^{-1}} = \zeta\frac{S_m^{3/2}(Ri)}{S_h(Ri)} \implies \boxed{\zeta = Ri\,\frac{S_h(Ri)}{S_m^{3/2}(Ri)}} \tag{4}$$

Note on Correction: Equation (4) corrects the power on S_m from 1 to $3/2$. This adjustment ensures internal self-consistency across all $Pr \neq 1$ formulations.

2 Stable Boundary Layer Formulation ($\zeta \geq 0$)

We generalize the stable non-dimensional profiles by retaining a linear momentum function ($\alpha_m = 1$) while allowing a general exponent α_h for heat:

$$\phi_m(\zeta) = 1 + \beta_m\zeta, \qquad \phi_h(\zeta) = (1+\beta_h\zeta)^{\alpha_h} \qquad (\zeta \geq 0) \tag{5}$$

Defining $x \equiv S_m^{1/2}(Ri) = \phi_m^{-1}(\zeta)$, we invert $\phi_m(\zeta)$ to obtain:

$$\zeta = \frac{1-x}{\beta_m x} \tag{6}$$

2.1 Master Implicit Equation for $x \equiv S_m^{1/2}$

Substituting (6) into (4) gives the explicit expression for $S_h(Ri)$:

$$S_h(Ri) = \frac{x^2(1-x)}{\beta_m Ri} \tag{7}$$

Equating $\phi_h = x/S_h = \frac{\beta_m Ri}{x(1-x)}$ with $\phi_h(\zeta) = \left(1 + \beta_h \frac{1-x}{\beta_m x}\right)^{\alpha_h}$ produces the master implicit equation for arbitrary α_h:

$$\boxed{\left[\frac{Ri\,\beta_m}{x(1-x)}\right]^{1/\alpha_h} = \frac{\beta_m x + \beta_h(1-x)}{\beta_m x}} \tag{8}$$

2.2 Case 1: Standard Linear Heat Profile ($\alpha_h = 1$)

Setting $\alpha_h = 1$ reduces (8) directly to a quadratic polynomial in x:

$$Ri(\beta_h - \beta_m)x^2 - (1 + Ri\,\beta_h)x + 1 = 0 \tag{9}$$

Taking the physically meaningful root ($x \to 1$ as $Ri \to 0$) yields the exact closed-form momentum stability function:

$$\boxed{S_m(Ri) = x^2 = \left[\frac{(1 + Ri\,\beta_h) - \sqrt{(1 + Ri\,\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)}\right]^2} \tag{10}$$

The corresponding heat function $S_h(Ri)$ is evaluated directly via (7).

2.3 Case 2: Businger–Dyer Quadratic Heat Profile ($\alpha_h = 2$)

Setting $\alpha_h = 2$ and squaring both sides of (8) clears fractional exponents, yielding a cubic polynomial $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$:

$$\begin{aligned} a_3 &= -(\beta_m - \beta_h)^2 \\ a_2 &= (\beta_m - \beta_h)^2 - 2\beta_h(\beta_m - \beta_h) \\ a_1 &= 2\beta_h(\beta_m - \beta_h) - \beta_h^2 - Ri\,\beta_m^3 \\ a_0 &= \beta_h^2 \end{aligned} \tag{11}$$

Special Equal-Parameter Limit ($\beta_m = \beta_h = \beta$):
When $\beta_m = \beta_h = \beta$, $a_3 = 0$ and $a_2 = 0$. The cubic collapses to $a_1x + a_0 = 0$:

$$x = \frac{1}{1 + \beta Ri} \implies \boxed{S_m(Ri) = (1 + \beta Ri)^{-2}, \qquad S_h(Ri) = (1 + \beta Ri)^{-3}} \tag{12}$$

3 Unstable Boundary Layer Formulation ($\zeta < 0$)

For unstable stratification ($\zeta < 0$), standard Businger–Dyer profile functions take the generalized exponential form:

$$\phi_m(\zeta) = (1 - \beta_m\zeta)^{-\alpha_m}, \qquad \phi_h(\zeta) = (1 - \beta_h\zeta)^{-\alpha_h} \tag{13}$$

For classical coefficients ($\alpha_m = 1/4, \alpha_h = 1/2$), the stability functions map in terms of ζ as:

$$S_m(\zeta) = (1 - \beta_m \zeta)^{1/2} \tag{14}$$

$$S_h(\zeta) = (1 - \beta_h \zeta)^{1/2}(1 - \beta_m \zeta)^{1/4} \tag{15}$$

3.1 Direct Coupling & Master Cubic in $x \equiv S_m^2$

Inverting (14) gives $1 - \beta_m \zeta = S_m^2$, allowing S_h to be expressed directly in terms of S_m:

$$S_h(S_m) = S_m^{1/2} \left[1 - \frac{\beta_h}{\beta_m}(1 - S_m^2) \right]^{1/2} \tag{16}$$

Substituting (16) into the corrected transformation $\zeta^2 = Ri^2 \frac{S_h^2}{S_m^3}$ and defining $x \equiv S_m^2(Ri)$ yields the exact master cubic equation for unstable stratification:

$$\boxed{x^3 - 2x^2 + \left(1 - Ri^2 \beta_m \beta_h\right) x - Ri^2 \beta_m (\beta_m - \beta_h) = 0} \tag{17}$$

3.2 Analytical Solution for Equal Coefficients ($\beta_m = \beta_h = 16$)

When $\beta_m = \beta_h = 16$, the constant term in (17) vanishes. Factoring out $x \neq 0$:

$$x^2 - 2x + (1 - 256 Ri^2) = 0 \implies (1 - x)^2 = 256 Ri^2 \implies x = 1 - 16 Ri \tag{18}$$

Recalling $x \equiv S_m^2$, we recover the analytical expression:

$$\boxed{S_m(Ri) = (1 - 16 Ri)^{1/2}} \tag{19}$$

4 Series Reversions & Pedagogical Models

4.1 Near-Neutral Series Expansions via Series Reversion

When exact inversion is mathematically intractable, Taylor expanding $\phi_m(\zeta) = 1 + \beta_m \zeta + \gamma_m \zeta^2$ and $\phi_h(\zeta) = 1 + \beta_h \zeta + \gamma_h \zeta^2$ around $\zeta = 0$ provides a forward series for $Ri(\zeta)$:

$$Ri(\zeta) = \zeta + (\beta_h - 2\beta_m)\zeta^2 + (3\beta_m^2 - 2\beta_m \beta_h - 2\gamma_m + \gamma_h)\zeta^3 + O(\zeta^4) \tag{20}$$

Reverting this series yields the inverted expression for $\zeta(Ri)$:

$$\boxed{\zeta(Ri) = Ri + (2\beta_m - \beta_h) Ri^2 + (5\beta_m^2 - 6\beta_m \beta_h + 2\beta_h^2) Ri^3 + O(Ri^4)} \tag{21}$$

Substituting $\zeta(Ri)$ into $S_m = \phi_m^{-2}$ and $S_h = (\phi_m \phi_h)^{-1}$ yields the low-order Ri-series forms:

$$S_m(Ri) = 1 - 2\beta_m Ri - \beta_m(\beta_m - 2\beta_h)Ri^2 + O(Ri^3) \tag{22}$$

$$S_h(Ri) = 1 - (\beta_m + \beta_h)Ri - (\beta_m^2 - 2\beta_h^2 + \gamma_m + \gamma_h)Ri^2 + O(Ri^3) \tag{23}$$

4.2 One-Parameter Toy Model with Finite Ri_c

Under the explicit assumption of equal profile parameters ($\beta_m = \beta_h = \beta \implies Pr_t = 1$), the transformation reduces to $Ri = \frac{\zeta}{1+\beta\zeta}$. Inverting directly gives:

$$\zeta(Ri) = \frac{Ri}{1 - \beta Ri} \implies \phi_m(Ri) = \frac{1}{1 - \beta Ri} \tag{24}$$

Defining the critical Richardson number $Ri_c \equiv 1/\beta$, the closure functions reduce to:

$$\boxed{S_m(Ri) = S_h(Ri) = \left(1 - \frac{Ri}{Ri_c}\right)^2 = \left(\frac{Ri_c - Ri}{Ri_c}\right)^2} \tag{25}$$

5 Surface Layer Roughness (z_0) Integration

In numerical boundary layer models with finite vertical grid resolution, evaluating Ri across the lowest model layer (z_1) requires explicit integration of surface roughness z_0. The bulk layer Richardson number is defined using effective integrated depth z_{eff}:

$$Ri_b = \frac{g}{\theta_0} \frac{\Delta\theta \, \Delta z}{(\Delta u)^2} \tag{26}$$

To ensure equivalence between discrete Ri_b closures and continuous MOST profiles, the effective height ratio ζ_{eff} is evaluated using logarithmic profile transformations:

$$z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_0) - \Psi_m(z_1/L)}{\ln(z_1/z_0) - \Psi_h(z_1/L)} \right] \tag{27}$$

This formulation preserves consistency between boundary-layer stability functions and surface flux routines.

6 Contrast of Formulations & Surface-Layer Discretization

While the gradient Richardson number (Ri) formulation is theoretically equivalent to the dimensionless height (ζ) formulation, numerical boundary-layer models require discrete treatment at the lowest model level (z_1). In first-order closures, lowest-level surface fluxes are linked to mean gradients by:

$$K_m \frac{\partial u}{\partial z} = u_*^2 \cos\alpha, \tag{28}$$

$$K_m \frac{\partial v}{\partial z} = u_*^2 \sin\alpha, \tag{29}$$

$$K_h \frac{\partial \theta}{\partial z} = u_* \theta_* \tag{30}$$

where $\tan\alpha = (\partial v/\partial z)/(\partial u/\partial z)$. Expressed separately for momentum (z_{0m}) and heat/scalars (z_{0h}), standard MOST determines friction parameters via path integrals:

$$u_* = \frac{\kappa|\mathbf{V}_1|}{\ln\left(\frac{z_1}{z_{0m}}\right) - \psi_m(\zeta_1)}, \qquad \psi_m(\zeta) = \int_0^{\zeta} \frac{1-\phi_m(\xi)}{\xi}\, d\xi, \tag{31}$$

$$\theta_* = \frac{\kappa(\theta_1 - \theta_g)}{\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1} - \psi_h(\zeta_1)}, \qquad \psi_h(\zeta) = \int_0^{\zeta} \frac{1-\phi_h(\xi)}{\xi}\, d\xi \tag{32}$$

where z_1 is the first mid-layer height, θ_g is the surface skin temperature, and $kB^{-1} \equiv \ln(z_{0m}/z_{0h})$ parameterizes excess thermal resistance at the microscale.

Evaluating (**??**)–(**??**) requires numerical iteration because $\zeta_1 = z_1/L$ depends non-linearly on u_* and θ_*. This computational bottleneck is eliminated in a closed-form *Ri* formulation.

6.1 Dual Logarithmic Discretization & Revised Scale h_1

Applying centered logarithmic finite differences $\frac{\partial}{\partial z} = \frac{1}{z}\frac{\partial}{\partial \ln(z/z_0)}$ across the surface layer yields direct flux relations:

$$u_* = \sqrt{S_m(Ri_{1/2})}\frac{\kappa|\mathbf{V}_1|}{\ln\left(\frac{z_1}{z_{0m}}\right)}, \tag{33}$$

$$\theta_* = \frac{S_h(Ri_{1/2})}{\sqrt{S_m(Ri_{1/2})}} \frac{\kappa(\theta_1 - \theta_g)}{\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1}} \tag{34}$$

To maintain consistency when $z_{0m} \neq z_{0h}$, the layer Richardson number $Ri_{1/2}$ incorporates separate momentum and thermal logarithmic profiles:

$$Ri_{1/2} = \frac{g}{\theta_0} \frac{\frac{1}{z_{\mathrm{mid},h}} \frac{\theta_1 - \theta_g}{\ln(z_1/z_{0h})}}{\frac{1}{z_{\mathrm{mid},m}^2} \frac{|\mathbf{V}_1|^2}{[\ln(z_1/z_{0m})]^2}} = \frac{g}{\theta_0} h_1 \frac{\theta_1 - \theta_g}{|\mathbf{V}_1|^2} \tag{35}$$

Evaluating vertical midpoints as geometric mean heights ($z_{\mathrm{mid},m} = \sqrt{z_1 z_{0m}} = \bar{z}_m$ and $z_{\mathrm{mid},h} = \sqrt{z_1 z_{0h}} =$

$\bar{z}_m e^{-\frac{1}{2}kB^{-1}}$) yields the explicit scale h_1:

$$\boxed{h_1 = \bar{z}_m \, e^{\frac{1}{2}kB^{-1}} \, \frac{\left[\ln\left(\frac{z_1}{z_{0m}}\right)\right]^2}{\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1}}} \tag{36}$$

6.2 Non-Iterative Surface Flux Formulation

Combining (**??**)–(**??**) gives closed-form surface exchange expressions:

$$u_*^2 = \frac{\kappa^2 \, S_m(Ri_{1/2}) \, |\mathbf{V}_1|^2}{\left[\ln\left(\frac{z_1}{z_{0m}}\right)\right]^2}, \tag{37}$$

$$u_*\theta_* = \frac{\kappa^2 \, S_h(Ri_{1/2}) \, |\mathbf{V}_1| \, (\theta_1 - \theta_g)}{\ln\left(\frac{z_1}{z_{0m}}\right) \left[\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1}\right]} \tag{38}$$

Equations (**??**)–(**??**) allow direct evaluation of surface stress and sensible heat flux in grid-scale boundary layer models without non-linear solver iterations.

7 Summary Reference of Core Equations

Table 1: Key Mathematical Equations for MOST $\leftrightarrow$ *Ri* Conversions

Regime / Application	**Equation**	**Mathematical Form**
General Linkage	Corrected $\zeta(Ri)$	$\zeta = Ri \, \frac{S_h(Ri)}{S_m^{3/2}(Ri)}$
Stable Master (α_h general)	Implicit $x = S_m^{1/2}$	$\left[\frac{Ri\,\beta_m}{x(1-x)}\right]^{1/\alpha_h} = \frac{\beta_m x + \beta_h(1-x)}{\beta_m x}$
Stable Linear ($\alpha_h = 1$)	Closed $S_m(Ri)$	$S_m(Ri) = \left[\frac{(1 + Ri\beta_h) - \sqrt{(1 + Ri\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)}\right]^2$
Unstable Master ($\alpha_m = \frac{1}{4}, \alpha_h = \frac{1}{2}$)	Master Cubic ($x \equiv S_m^2$)	$x^3 - 2x^2 + (1 - Ri^2\beta_m\beta_h)x - Ri^2\beta_m(\beta_m - \beta_h) = 0$
Unstable Equal ($\beta_m = \beta_h = 16$)	Closed $S_m(Ri)$	$S_m(Ri) = (1 - 16Ri)^{1/2}$
Equal-Profile Toy Model	$Pr_t = 1, Ri_c = 1/\beta$	$S_m(Ri) = S_h(Ri) = \left(1 - \frac{Ri}{Ri_c}\right)^2$